Network Layer: Data Plane

General Resources:

- NetworkingDataPlaneDefinitions.pdf
- NetworkingDataPlaneEquations.pdf

Topics Covered:

- Network layer: overview
  - data plane
  - control plane
- What's inside a router
  - input ports, switching, output ports
  - buffer management, scheduling
- IP: the Internet Protocol
  - datagram format
  - Addressing
  - network address translation
  - IPv6

Practice Questions: Multiple Choice

Question 1

Check all of the statements below about where (in the network) the network layer is implemented that are true.

- ☐ The network layer is implemented in Ethernet switches in a local area network.
- ☑ The network layer is implemented in routers in the network core.
- ☑ The network layer is implemented in hosts at the network's edge.
- ☐ The network layer is implemented in wired Internet-connected devices but not wireless Internet-connected devices.

Question 2

Consider the travel analogy discussed in the textbook - some actions we take on a trip correspond to forwarding and other actions we take on a trip correspond to routing. Which of the following travel actions below correspond to forwarding? The other travel actions that you don't select below then correspond to routing.

- ☑ A car stops at an intersection to “gas-up” and take a “bathroom break
- ☑ A car waits at light and then turns left at the intersection.
- ☑ A car takes the 3rd exit from a roundabout.
- ☐ A traveler decides to fly to Sydney through Singapore rather that Dubai.
- ☐ A car takes highway 80 between New York and Chicago, rather than highway 87 to Albany and from there take Interstate 90 to Chicago.
- ☐ A climber decides to take the South Col Route to the top of Mt Everest rather than the Northeast Ridge route.

Question 3

For each of the actions below, select those actions below that are primarily in the network-layer data plane. The other actions that you don't select below then correspond to control-plane actions.

- ☐ Monitoring and managing the configuration and performance of an network device.
- ☑ Looking up address bits in an arriving datagram header in the forwarding table.
- ☑ Moving an arriving datagram from a router’s input port to output port
- ☐ Computing the contents of the forwarding table.
- ☑ Dropping a datagram due to a congested (full) output buffer.

Question 4

We've seen that there are two approaches towards implementing the network control plane - a per-router control-plane approach and a software-defined networking (SDN) control-plane approach. Which of the following actions occur in a per-router control-plane approach?

The other actions that you don't select below then correspond to actions in an SDN control plane.

- ☐ A control agent in router receives a complete forwarding table, which it installs and uses to locally control datagram forwarding
- ☑ A router exchanges messages with another router, indicating the cost for it (the sending router) to reach a destination host.
- ☐ All routers in the network send information about their incoming and outgoing links to a logically centralized controller.
- ☑ Routers send information about their incoming and outgoing links to other routers in the network.

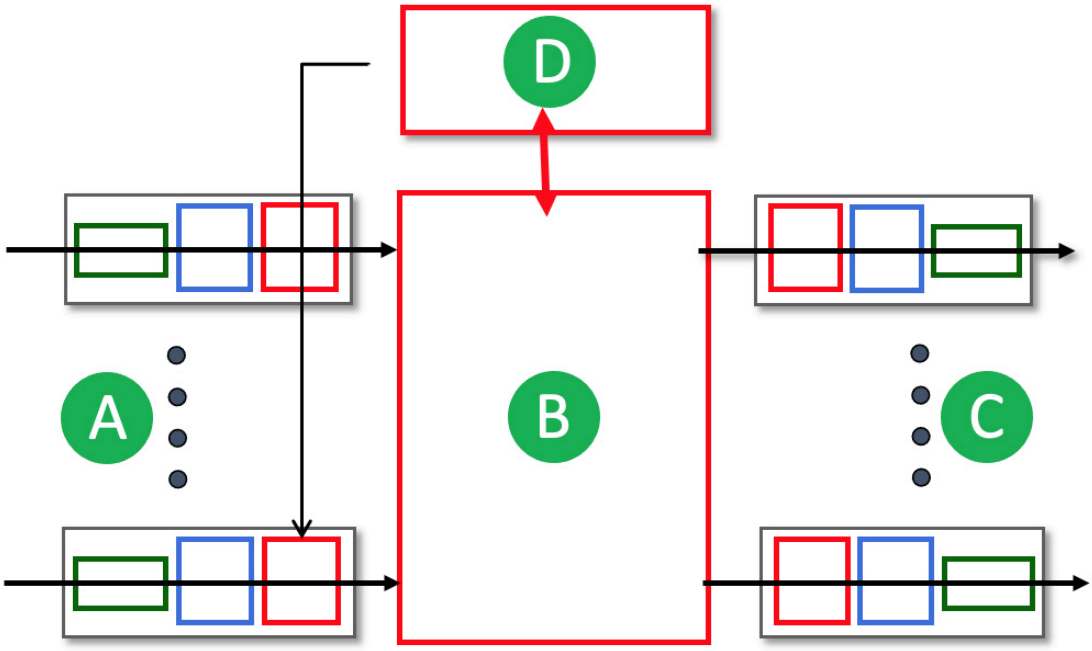
Question 5

Which of the following quality-of-service guarantees are part of the Internet's best-effort service model? Check all that apply.

- ☐ A guaranteed minimum bandwidth is provided to a source-to-destination flow of packets
- ☐ In-order datagram payload delivery to the transport layer of those datagrams arriving to the receiving host.
- ☐ Guaranteed delivery from sending host to receiving host.
- ☐ Guaranteed delivery time from sending host to receiving host.
- ☑ None of the other services listed here are part of the best-effort service model. Evidently, best-effort service really means no guarantees at all!

Question 6

Match the names of the principal router components (A,B,C,D below) with their function and whether they are in the network-layer data plane or control plane.



Answer List:

- input ports, operating primarily in the data plane.
- the switching fabric, operating primarily in the control plane.
- the routing processor, operating primarily in the data plane.
- input ports, operating primarily in the control plane.
- the routing processor, operating primarily in the control plane.
- output ports, operating primarily in the data plane.
- output ports, operating primarily in the control plane.
- the switching fabric, operating primarily in the data plane.

Question List:

- (A) are … input ports, operating primarily in the data plane.
- (B) is … the switching fabric, operating primarily in the data plane.
- (C) are … output ports, operating primarily in the data plane.
- (D) is … the routing processor, operating primarily in the control plane.

Question 7

Where in a router is the destination IP address looked up in a forwarding table to determine the appropriate output port to which the datagram should be directed?

- ☐ Within the routing processor.
- ☑ At the input port where a packet arrives.
- ☐ Within the switching fabric.
- ☐ At the output port leading to the next hop towards the destination.

Question 8

Where in a router does "match plus action" happen to determine the appropriate output port to which the arriving datagram should be directed?

- ☐ Within the switching fabric.
- ☐ At the output port leading to the next hop towards the destination.
- ☐ Within the routing processor.
- ☑ At the input port where a packet arrives.

Question 9

Consider the following forwarding table below. Indicate the output to link interface to which a datagram with the destination addresses below will be forwarded under longest prefix matching. (Note: The list of addresses is ordered below. If two addresses map to the same output link interface, map the first of these two addresses to the first instance of that link interface.) [Note: You can find more examples of problems similar to this here.]

Destination Address Range	Link interface
11001000 00010111 00010*** ********	0
11001000 00010111 00011000 ********	1
11001000 00010111 00011*** ********	2
otherwise	3

Answer List:

- This is the second destination address in the list that maps to output port 2.
- This is the second destination address in the list that maps to output port 3.
- This is the first destination address in the list that maps to output port 0.
- This is the first destination address in the list that maps to output port 3.
- This is the first destination address in the list that maps to output port 1.
- This is the second destination address in the list that maps to output port 0.
- This is the second destination address in the list that maps to output port 1.
- This is the first destination address in the list that maps to output port 2.

Question List:

- 11001000 00010111 00010010 10101101: **This is the first destination address in the list that maps to output port 0.**
- 11001000 00010111 00011000 00001101: **This is the first destination address in the list that maps to output port 1.**
- 11001000 00010111 00011001 11001101: **This is the first destination address in the list that maps to output port 2.**
- 10001000 11100000 00011000 00001101: **This is the first destination address in the list that maps to output port 3.**

- 11001000 00010111 00011000 11001111: **This is the second destination address in the list that maps to output port 1.**
- 11001000 00010111 00010001 01010101: **This is the second destination address in the list that maps to output port 0.**
- 11001000 00010111 00011101 01101101: **This is the second destination address in the list that maps to output port 2.**

Question 10

Suppose a datagram is switched through the switching fabric and arrives to its appropriate output to find that there are no free buffers. In this case:

- ☐ Another packet will be removed (lost) from the buffer to make room for this packet.
- ☐ The packet will be sent back to the input port.
- ☐ The packet will be dropped (lost).
- ☑ The packet will either be dropped or another packet will be removed (lost) from the buffer to make room for this packet, depending on policy. But the packet will definitely not be be sent back to the input port.

Question 10

What is meant by Head of the Line (HOL) blocking?

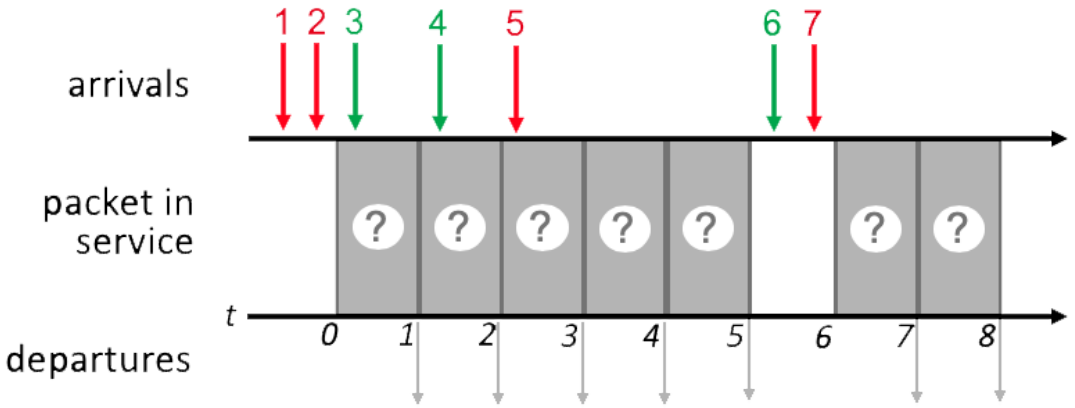
- ☑ A queued datagram waiting for service at the front of a queue prevents other datagrams in queue from moving forward in the queue.
- ☐ In a block error code, the first bytes of the code indicate the type of coding being used.
- ☐ A queued datagram receiving service at the front of a queue prevents other datagrams in queue from receiving service.

Question 11

Consider the pattern of red and green packet arrivals to a router's output port queue, shown below. Suppose each packet takes one time slot to be transmitted, and can only begin transmission at the beginning of a time slot after its arrival. Indicate the sequence of departing packet numbers (at t = 1, 2, 3, 4, 5, 7, 8) under FCFS

scheduling. Give your answer as 7 ordered digits (each corresponding to the packet number of a departing packet), with a single space between each digit, and no spaces before the first or after the last digit, e.g., in a form like 7 6 5 4 3 2 1).

[Note: You can find more examples of problems similar to this here.]



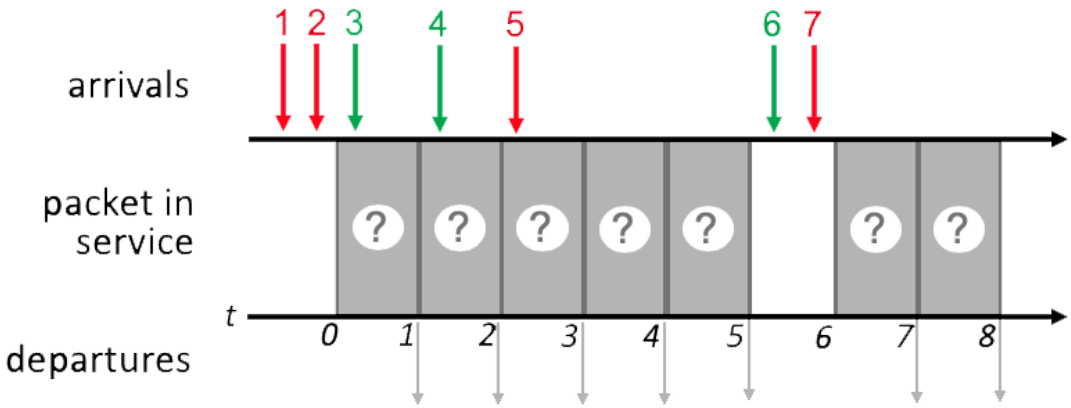
Answer: 1 2 3 4 5 6 7

Question 12

Consider the pattern of red and green packet arrivals to a router's output port queue, shown below. Suppose each packet takes one time slot to be transmitted, and can only begin transmission at the beginning of a time slot after its arrival. Indicate the sequence of departing packet numbers (at t = 1, 2, 3, 4, 5, 7, 8) under priority scheduling, where red packets have higher priority.

Give your answer as 7 ordered digits (each corresponding to the packet number of a departing packet), with a single space between each digit, and no spaces before the first or after the last digit, e.g., in a form like 7 6 5 4 3 2 1).

[Note: You can find more examples of problems similar to this here.]

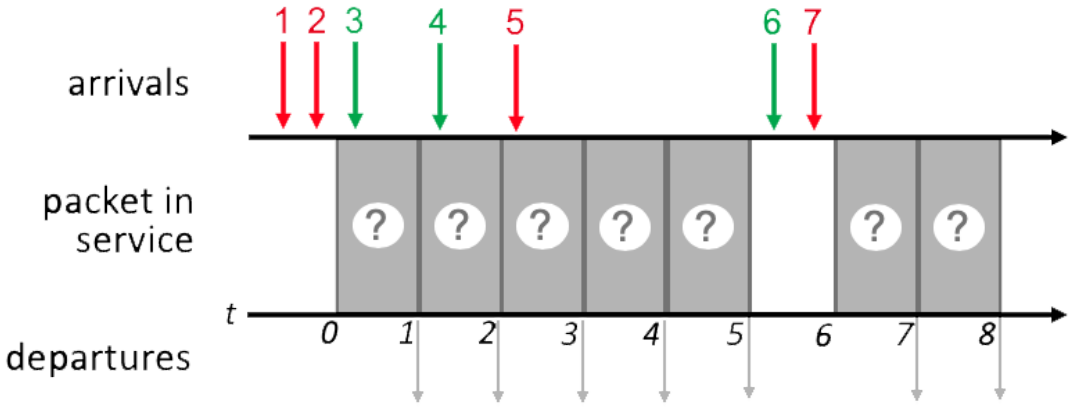


Answer: 1 2 3 5 4 7 6

Question 13

Consider the pattern of red and green packet arrivals to a router's output port queue, shown below. Suppose each packet takes one time slot to be transmitted, and can only begin transmission at the beginning of a time slot after its arrival. Indicate the sequence of departing packet numbers (at t = 1, 2, 3, 4, 5, 7, 8) under round robin scheduling, where red starts a round if there are both red and green packets ready to transmit after an empty slot.

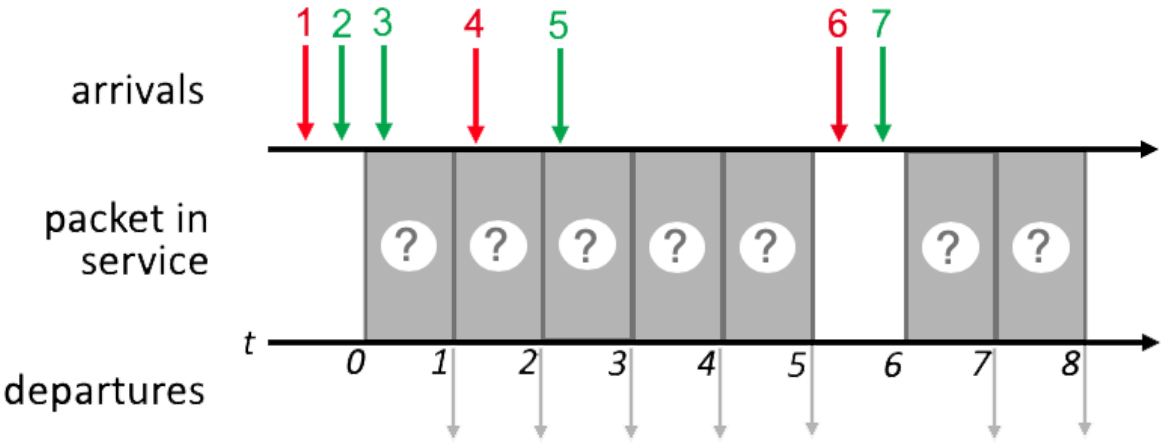
Give your answer as 7 ordered digits (each corresponding to the packet number of a departing packet), with a single space between each digit, and no spaces before the first or after the last digit, e.g., in a form like 7 6 5 4 3 2 1).



Answer: 1 3 2 4 5 7 6

Question 14

Consider the pattern of red and green packet arrivals to a router's output port queue, shown below. Suppose each packet takes one time slot to be transmitted, and can only begin transmission at the beginning of a time slot after its arrival. Indicate the sequence of departing packet numbers (at t = 1, 2, 3, 4, 5, 7, 8) under FCFS scheduling. Give your answer as 7 ordered digits (each corresponding to the packet number of a departing packet), with a single space between each digit, and no spaces before the first or after the last digit, e.g., in a form like 7 6 5 4 3 2 1).

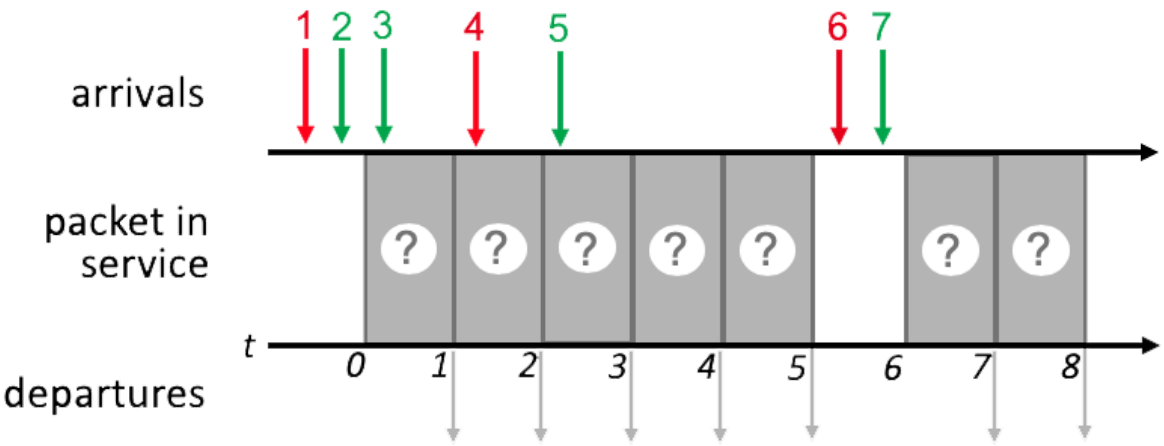


Answer: 1 2 3 4 5 6 7

Question 15

Consider the pattern of red and green packet arrivals to a router's output port queue, shown below. Suppose each packet takes one time slot to be transmitted, and can only begin transmission at the beginning of a time slot after its arrival. Indicate the sequence of departing packet numbers (at t = 1, 2, 3, 4, 5, 7, 8) under priority scheduling, where red packets have higher priority.

Give your answer as 7 ordered digits (each corresponding to the packet number of a departing packet), with a single space between each digit, and no spaces before the first or after the last digit, e.g., in a form like 7 6 5 4 3 2 1).

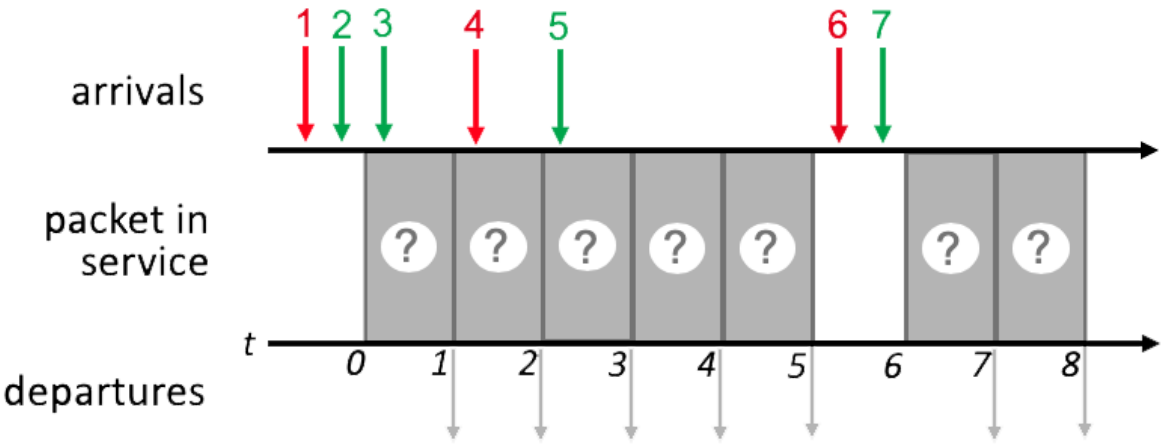


Answer: 1 2 4 3 5 6 7

Question 16

Consider the pattern of red and green packet arrivals to a router's output port queue, shown below. Suppose each packet takes one time slot to be transmitted, and can only begin transmission at the beginning of a time slot after its arrival. Indicate the sequence of departing packet numbers (at t = 1, 2, 3, 4, 5, 7, 8) under round robin scheduling, where red starts a round if there are both red and green packets ready to transmit after an empty slot.

Give your answer as 7 ordered digits (each corresponding to the packet number of a departing packet), with a single space between each digit, and no spaces before the first or after the last digit, e.g., in a form like 7 6 5 4 3 2 1).



Answer: 1 2 4 3 5 6 7

Question 17

What are the principal components of the IPv4 protocol (check all that apply)?

- ☑ IPv4 datagram format.
- ☐ Routing algorithms and protocols like OSPF and BGP.
- ☐ ICMP (Internet Control Message Protocol)
- ☑ IPv4 addressing conventions.
- ☑ Packet handling conventions at routers (e.g., segmentation/reassembly)
- ☐ SDN controller protocols.

Question 18

Match each of the following fields in the IP header with its description, function or use.

Answer List:

- This field contains the "protocol number" for the transport-layer protocol to which this datagram's payload will be demultiplexed - UDP or TCP, for example.
- This field is used for datagram fragmentation/reassembly.
- The value in this field is decremented at each router; when it reaches zero, the packet must be dropped.
- This field contains the IP protocol version number.
- This field contains a UDP or TCP segment, for example.
- This field contains ECN and differentiated service bits.
- This field contains the Internet checksum of this datagram's header fields.
- This field indicates the total number of bytes in datagram.

Question List:

- Version field: This field contains the IP protocol version number.
- Type-of-service field: This field contains ECN and differentiated service bits.
- Fragmentation offset field: This field is used for datagram fragmentation/reassembly.
- Time-to-live field: The value in this field is decremented at each router; when it reaches zero, the packet must be dropped.
- Header checksum field: This field contains the Internet checksum of this datagram's header fields.
- Upper layer field: This field contains the "protocol number" for the transport-layer protocol to which this datagram's payload will be demultiplexed - UDP or TCP, for example.
- Payload/data field: This field contains a UDP or TCP segment, for example.
- Datagram length field: This field indicates the total number of bytes in datagram.

Question 19

Which of the following statements is true regarding an IP address? (Zero, one or more of the following statements is true).

- ☐ It is not necessary for a device using the IP protocol to actually have an IP address associated with it.

- ☑ An IP address is associated with an interface.
- ☑ If a host has more than one interface, then it has more that one IP address at which it can be reached.
- ☑ If a router has more than one interface, then it has more that one IP address at which it can be reached.

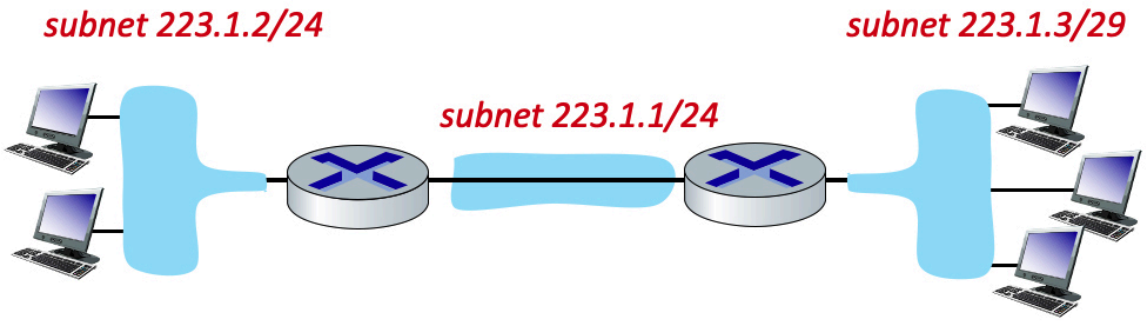
Question 20

What is meant by an IP subnet? (Check zero, one or more of the following characteristics of an IP subnet).

- ☑ A set of device interfaces that can physically reach each other without passing through an intervening router.
- ☐ A set of devices all manufactured by the same equipment maker/vendor.
- ☑ A set of devices that have a common set of leading high order bits in their IP address.
- ☐ A set of devices that always have a common first 16 bits in their IP address.

Question 21

Consider the three subnets in the diagram below.



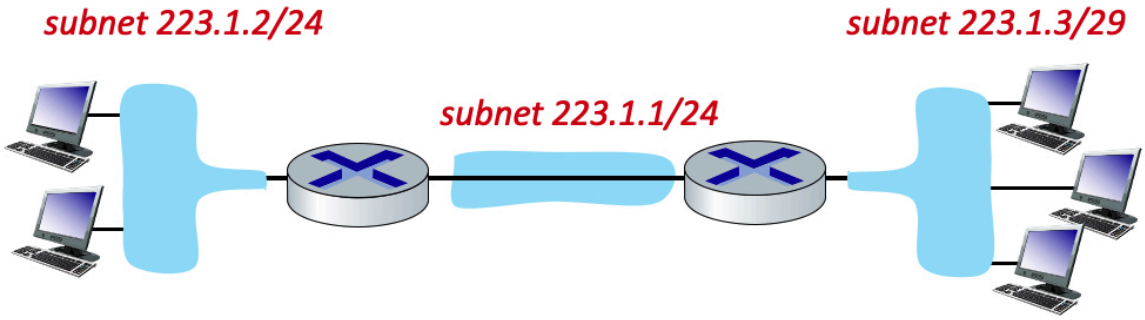
What is the maximum # of interfaces in the 223.1.2/24 network?

- ☐ 2**32
- ☑ 256
- ☐ There's no a priori limit on the number of interfaces in this subnet.

- [ ] 128
- [ ] Two hosts, as shown in the figure.

Question 22

Consider the three subnets in the diagram below.

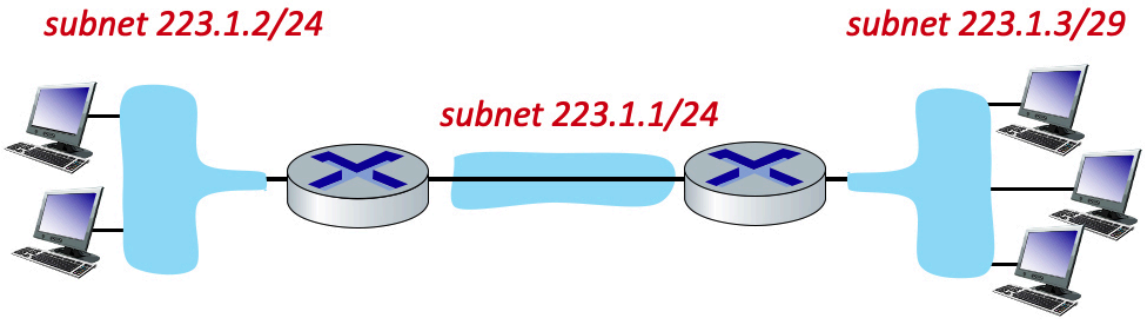


What is the maximum # of interfaces in the 223.1.3/29 network?

- [ ] 2**32
- [x] 8
- [ ] 128
- [ ] There's no a priori limit on the number of interfaces in this subnet.
- [ ] Three hosts, as shown in the figure.

Question 23

Consider the three subnets in the diagram below.



Which of the following addresses can not be used by an interface in the 223.1.3/29 network? Check all that apply.

- ☐ 223.1.3.2
- ☑ 223.1.3.28
- ☑ 223.1.3.16
- ☐ 223.1.3.6
- ☑ 223.1.2.6

Question 24

What is meant by saying that DHCP is a "plug and play" protocol?

- ☐ The network provides an Ethernet jack for a host’s Ethernet adapter.
- ☐ The host needs to “plug” (by wire or wirelessly) into the local network in order to access (“play” in) the Internet
- ☑ No manual configuration is needed for the host to join the network.

Question 25

Which of the following statements about a DHCP request message are true (check all that are true). Hint: check out Figure 4.24 in the 7th and 8th edition of our textbook.

- ☐ A DHCP request message is optional in the DHCP protocol.
- ☐ A DHCP request message is sent from a DHCP server to a DHCP client.
- ☑ The transaction ID in a DHCP request message will be used to associate this message with future DHCP messages sent from, or to, this client.
- ☐ The transaction ID in a DCHP request message is used to associate this message with previous messages sent by this client.
- ☑ A DHCP request message may contain the IP address that the client will use.
- ☑ A DHCP request message is sent broadcast, using the 255.255.255.255 IP destination address.

Question 26

Which of the following fields occur ONLY in the IPv6 datagram header (i.e., appear in the IPv6 header but not in the IPv4 header)? Check all that apply.

- ☐ The upper layer protocol (or next header) field.
- ☑ The flow label field.
- ☐ The IP version number field.
- ☐ The header checksum field.
- ☑ 128-bit source and destination IP addresses.
- ☐ The options field.
- ☐ The header length field.
- ☐ The time-to-live (or hop limit) field.

Question 27

What is the purpose of the Dynamic Host Configuration Protocol?

- ☐ To configure the set of available open ports (and hence well-known services) for a server.
- ☑ To obtain an IP address for a host attaching to an IP network.
- ☐ To get the 48-bit link-layer MAC address associated with a network-layer IP address.
- ☐ To configure the interface speed to be used, for hardware like Ethernet, which can be used at different speeds.

Question 28

Destination-based forwarding, which we studied in section 4.2, is a specific instance of match+action and generalized forwarding. Select the phrase below which best completes the following sentence:

"In destination-based forwarding, ..."

- ☐ ... after matching on the port number in the segment's header, the action taken is to decide whether or not to drop the datagram containing that segment.
- ☑ ... after matching on the destination IP address in the datagram header, the action taken is to forward the datagram to the output port associated with that destination IP address.

- ☐ ... after matching on the 48-bit link-layer destination MAC address, the action taken is to forward the datagram to the output port associated with that link-layer address.
- ☐ ... after matching on the source and destination IP address in the datagram header, the action taken is to forward the datagram to the output port associated with that source and destination IP address pair.
- ☐ ... after matching on the port number in the segment's header, the action taken is to forward the datagram to the output port associated with that port number.
- ☐ ... after matching on the destination IP address in the datagram header, the action taken is to decide whether or not to drop that datagram.
- ☐ ... after matching on the URL contained in an HTTP GET request in the TCP segment within the IP datagram, the action taken is to determine the IP address of the server associated with that URL, and to forward the datagram to the output port associated with that destination IP address.

Question 29

Which of the following match+actions can be taken in the generalized OpenFlow 1.0 match+action paradigm that we studied in Section 4.4? Check all that apply.

- ☑ ... after matching on the port number in the segment's header, the action taken is to decide whether or not to drop that datagram containing that segment.
- ☑ ... after matching on the destination IP address in the datagram header, the action taken is to decide whether or not to drop that datagram.
- ☑ ... after matching on the destination IP address in the datagram header, the action taken is to forward the datagram to the output port associated with that destination IP address.
- ☑ ... after matching on the port number in the segment's header, the action taken is to forward the datagram to the output port associated with that destination IP address.
- ☐ ... after matching on the URL contained in an HTTP GET request in the TCP segment within the IP datagram, the action taken is to

determine the IP address of the server associated with that URL, and to forward the datagram to the output port associated with that destination IP address.

- ☑ ... after matching on the source and destination IP address in the datagram header, the action taken is to forward the datagram to the output port associated with that source and destination IP address pair.
- ☑ ... after matching on the 48-bit link-layer destination MAC address, the action taken is to forward the datagram to the output port associated with that link-layer address.

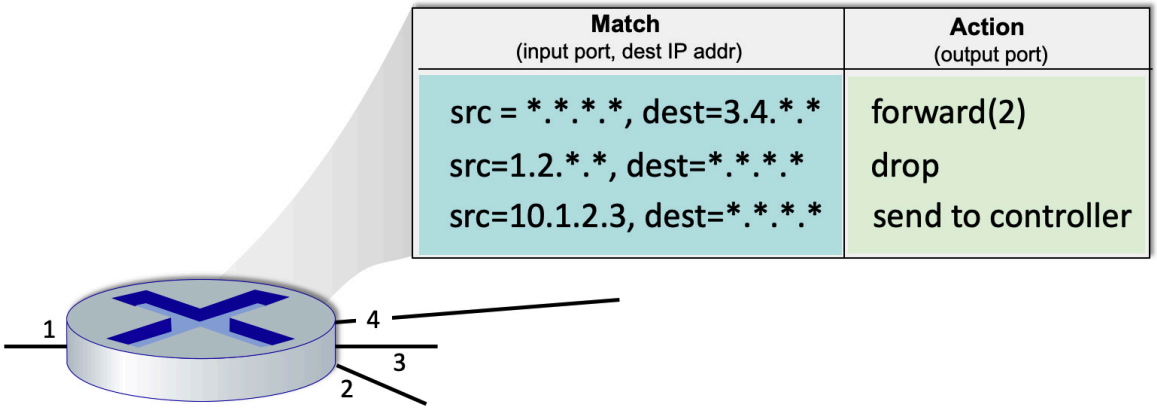
Question 30

Which of the following fields in the frame/datagram/segment/application-layer message can be matched in OpenFlow 1.0? Check all that apply.

- ☐ Time-to-live field
- ☑ IP type-of-service field
- ☑ Upper layer protocol field
- ☑ IP source address
- ☑ IP destination address
- ☐ Number of bytes in the datagram
- ☑ Source and/or destination port number
- ☐ URL in HTTP message

Question 31

Consider the figure below that shows the generalized forwarding table in a router. Recall that a * represents a wildcard value. Now consider an arriving datagram with the IP source and destination address fields indicated below. For each source/destination IP address pair, indicate which rule is matched. Note: assume that a rule that is earlier in the table takes priority over a rule that is later in the table and that a datagram that matches none of the table entries is dropped.



Answer List:

- No match to any rule.
- Rule 3, with action send to controller
- Rule 1, with action forward(2)
- Rule 2, with action drop

Question List:

- Source: 1.2.56.32 Destination:128.116.40.186: **Rule 2, with action drop**
- Source: 65.92.15.27 Destination: 3.4.65.76: **Rule 1, with action forward(2)**
- Source: 10.1.2.3 Destination: 7.8.9.2: **Rule 3, with action send to controller**
- Source: 10.1.34.56 Destination: 54.72.29.90: **No match to any rule.**

Question 32

Consider the network below. We want to specify the match+action rules at s3 so that only the following network-wide behavior is allowed:

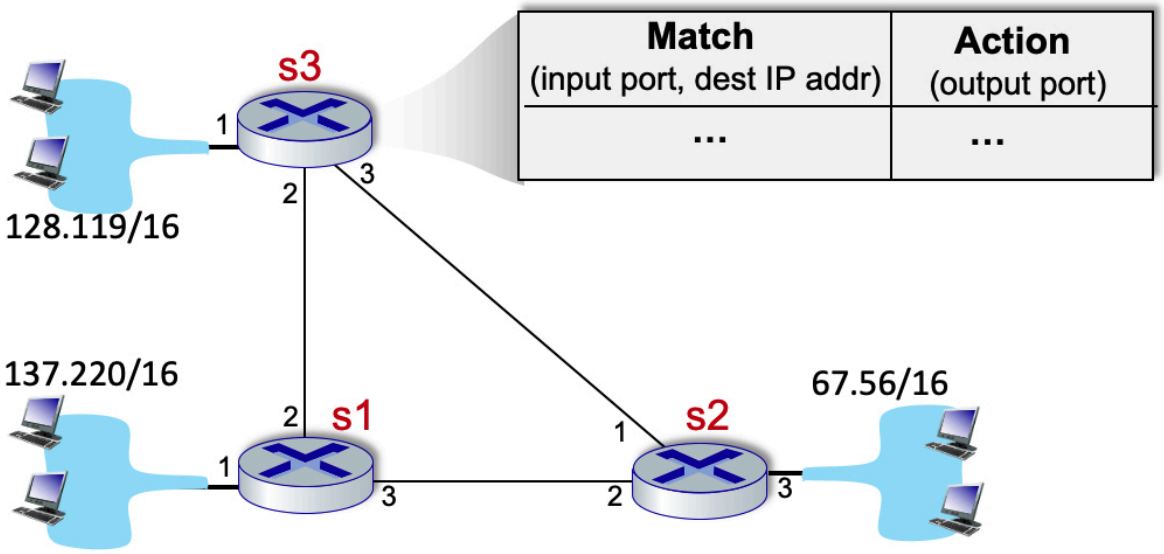
traffic from 128.119/16 and destined to 137.220/16 is forwarded on the direct link from s3 to s1;

traffic from 128.119/16 and destined to 67.56/16 is forwarded on the direct link from s3 to s2;

incoming traffic via port 2 or 3, and destined to 128.119/16 is forwarded to 128.119/16 via local port 1.

No other forwarding should be allowed. In particular s3 should not forward traffic arriving from 137.220/16 and destined for 67.56/16 and vice versa.

From the list of match+action rules below, select the rules to include in s3's flow table to implement this forwarding behavior. Assume that if a packet arrives and finds no matching rule, it is dropped.

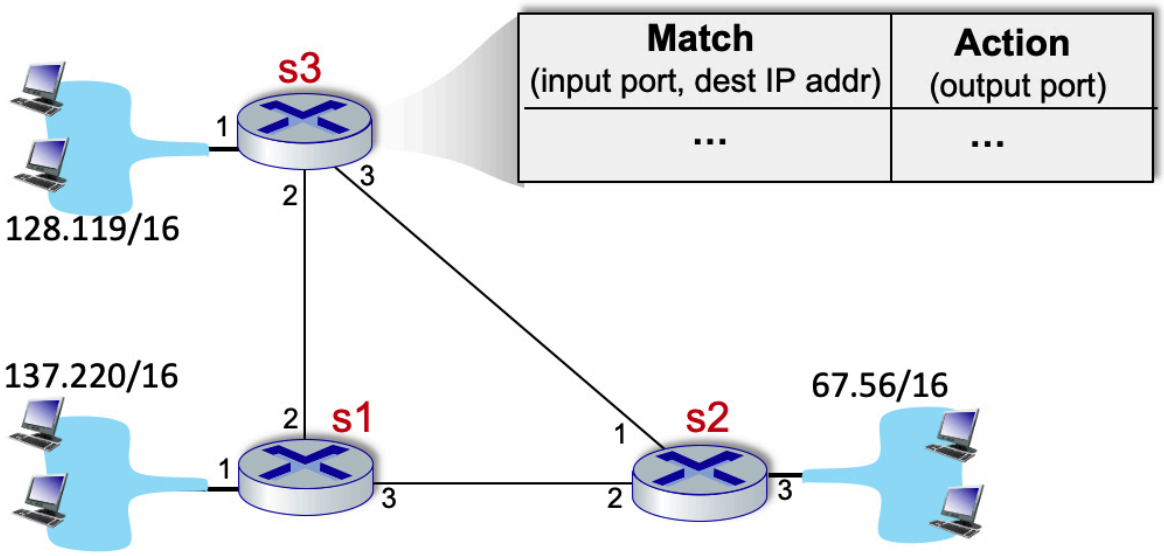


- [x] Input port: 3; Dest: 128.119/16 Action: forward(1)
- [x] Input port:1 ; Dest: 137.220/16 Action: forward(2)
- [ ] Input port:1 ; Dest: 137.220/16 Action: forward(3)
- [x] Input port: 2; Dest: 128.119/16 Action: forward(1)
- [ ] Input port: 3; Dest: 137.220/16 Action: forward(2)
- [ ] Input port: 2; Dest: 67.56/16 Action: forward(3)
- [ ] Input port: 1; Dest: 67.56/16 Action: forward(2)
- [x] Input port: 1; Dest: 67.56/16 Action: forward(3)

Question 33

Consider the network below. We want to specify the match+action rules at s3 so that s3 acts only as a relay for traffic between 137.220/16 and 67.56/16. In particular s3 should not accept/forward and traffic to/from 128.119/16.

From the list of match+action rules below, select the rules to include in s3's flow table to implement this forwarding behavior. Assume that if a packet arrives and finds no matching rule, it is dropped.



- [x] Input port: 3; Dest: 137.220/16 Action: forward(2)
- [ ] Input port: 1; Dest: 67.56/16 Action: forward(2)
- [ ] Input port:1 ; Dest: 137.220/16 Action: forward(3)
- [ ] Input port:1 ; Dest: 137.220/16 Action: forward(2)
- [ ] Input port: 3; Dest: 128.119/16 Action: forward(1)
- [x] Input port: 2; Dest: 67.56/16 Action: forward(3)
- [ ] Input port: 2; Dest: 128.119/16 Action: forward(1)
- [ ] Input port: 1; Dest: 67.56/16 Action: forward(3)

Question 34

What is meant by generalized forwarding (as opposed to destination-based forwarding) in a router or switch?

- [x] Any of several actions (including drop (block), forward to a given interface, or duplicate-and-forward) can be made based on the contents of one or more packet header fields.
- [ ] The decision about which output port to forward a packet to can be made based on the link-type of the outgoing port (e.g., Ethernet versus WiFi).
- [ ] In addition to performing forwarding, the device can generalize its services, also performing hop-by-hop reliable data transfer and per-hop congestion control.
- [ ] None of the other answers is a correct definition of generalized forwarding.

Question 35

Which of the following network devices can be thought of as a "middlebox"? Check all that apply.

- [ ] IP router
- [ ] WiFi base station
- [x] Network Address Translation box
- [x] HTTP cache
- [ ] SDN controller
- [x] HTTP load balancer

Question 36

What protocol (or protocols) constitutes the "thin waist" of the Internet protocol stack? Check all that apply.

- [ ] DNS
- [ ] TCP
- [ ] HTTP
- [ ] Ethernet
- [x] IP
- [ ] WiFi

Question 37

Which of the statements below are true statements regarding the "end-to-end principle"? Check all that apply.

- ☐ The end-to-end argument advocates placing functionality at the network edge to optimize performance, such as end-end delay.
- ☑ The end-to-end argument allows that some redundant functionality might be placed both in-network and at the network edge in order to enhance performance.
- ☑ The end-to-end argument advocates placing functionality at the network edge because some functionality cannot be completely and correctly implemented in the network, and so needs to be placed at the edge in any case, making in-network implementation redundant.

Question 38

What is meant when it is said that the Internet has an “hourglass” architecture? See the picture below if you are unfamiliar with an "hourglass".

- ☐ Packets flow from top to bottom down the stack, like sand in an hour glass. Then, on the receiver side, if the hourglass is reversed, packets flow up the stack, like sand flowing in the opposite direction.
- ☐ The Internet protocol stack has a “thin waist” in the middle, like an hourglass. The Internet Protocol (IP) is the only network-layer protocol in the middle layer of the stack. Every other layer has multiple protocols at that layer.
- ☐ ... after matching on the source and destination IP address in the datagram header, the action taken is to forward the datagram to the output port associated with that source and destination IP address pair.

Question 39

In the US, which of the following services has been regulated by the Federal Communications Commission (FCC) going back into the 20th century?

- ☐ Neither telecommunications services (broadly) nor information services; the FCC's jurisdiction is only on over-the-air (e.g., wireless) links.
- ☐ Information services.
- ☐ Both telecommunications services and information services
- ☑ Telecommunication services.